

HTML, CSS & JavaScript Basics — Assignment 03

Responsive CSS + JavaScript Fundamentals | Beginner Level

Topics Covered: CSS Grid, CSS Breakpoints, Media Queries, Responsive Design, DOM Manipulation, JavaScript Variables, Data Types

Important Instructions

- Read all questions carefully before starting.
- Create a **separate file for each question**. Do not combine all questions into one HTML or JavaScript file.
- Write clean, well-structured, and properly indented code.
- Use only the concepts taught in class.
- Do not use CSS or JavaScript frameworks/libraries such as Bootstrap, Tailwind, jQuery, etc.
- Test your code in the browser before submission.
- Your responsive webpage in Question 1 must work appropriately on both desktop/web and mobile screen sizes.

Submission Instructions

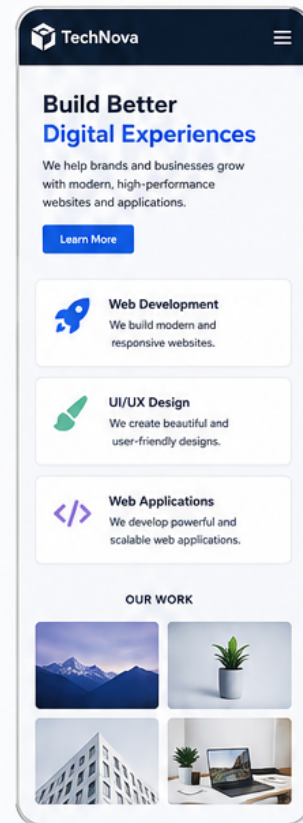
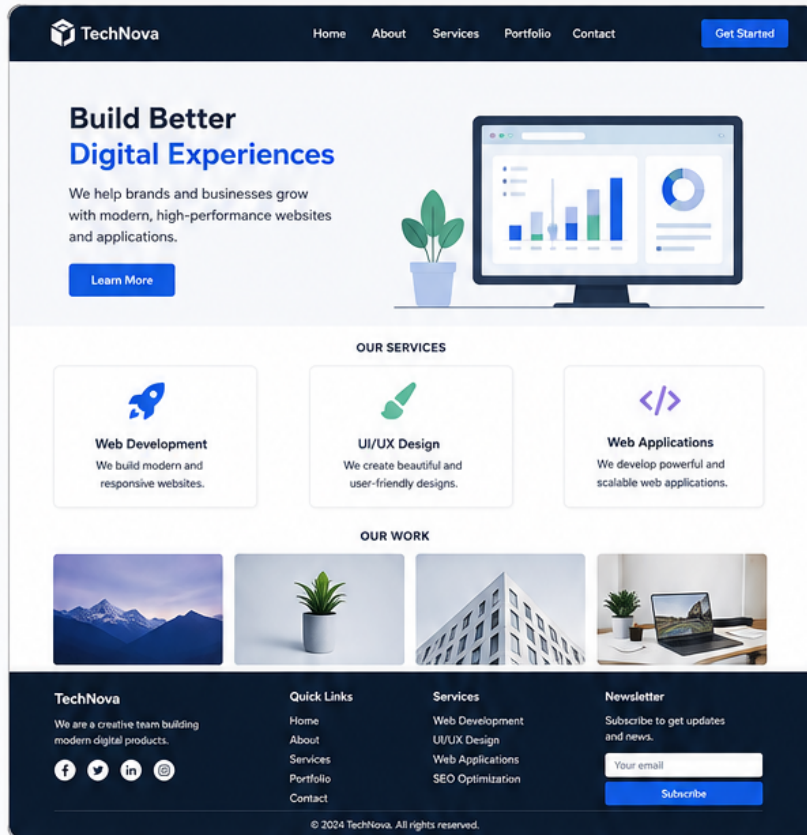
- Students must submit their assignment via a **link to their GitHub repository**.
- Create a **separate file for each question**.
- Make sure the GitHub repository contains all required HTML, CSS, and JavaScript files.
- Push your latest working code to GitHub before submitting the repository link.
- The repository should be accessible to the instructor.

Question 1 — Responsive Layout Using CSS Grid

Design the following webpage layout using **HTML and CSS Grid**. The design must be responsive for both desktop/web and mobile devices using appropriate **CSS breakpoints and media queries**.

Q1 Responsive Layout with CSS Grid

Design the following webpage layout using HTML and CSS Grid.
The layout must be fully responsive using breakpoints and media queries.



Requirements: Use CSS Grid for the layout. Make the design responsive for tablets and mobile devices. Use appropriate breakpoints and media queries. The layout should be clean and visually similar to the design shown above.

Requirements:

- Use **CSS Grid** for the main page layout.
- Create appropriate **breakpoints** for responsiveness.
- Use **media queries** to change the layout at different screen sizes.
- Make the design responsive for desktop, tablet, and mobile screens.
- On smaller screens, rearrange or stack the sections appropriately.
- Try to match the overall structure, spacing, typography, colors, and alignment of the reference design as closely as possible.
- You may use your own text, icons, and images, but the overall layout should remain similar to the reference.

Question 2 — DOM Manipulation with JavaScript

Create an HTML page and use JavaScript to demonstrate basic DOM manipulation using `document.write()` and `getElementById()`. You must also demonstrate the display methods `alert()` and `console.log()`.

HTML Requirements

- Create a heading with `id="title"`.
- Create a paragraph with `id="message"`.
- Create a button with `id="showBtn"`.

JavaScript Tasks

- Use **getElementById()** to access the heading and paragraph.
- When the button is clicked, display the heading text using **alert()**.
- Display the paragraph text using **console.log()**.
- Use **document.write()** somewhere in the page to display a simple message of your choice.

Suggested files: q2_dom.html and q2_script.js

Question 3 — JavaScript Variables & Data Types

Create an HTML page and a JavaScript file to demonstrate your understanding of JavaScript variables and data types.

Tasks

- Declare variables using **var**, **let**, and **const**.
- Create examples of JavaScript data types: **String**, **Number**, **Boolean**, **Undefined**, **Null**, **Object**, **Array**, **Symbol**, and **BigInt**.
- Use **console.log()** to display each variable and its value.
- Use **typeof** to display/check the type of your variables.
- Add short comments in your JavaScript code explaining what each data type represents with an example.

Suggested files: q3_variables.html and q3_variables.js

Final Submission Checklist

- Separate files created for Questions 1, 2, and 3.
- Question 1 is responsive on desktop and mobile.
- CSS Grid, breakpoints, and media queries are used in Question 1.
- DOM manipulation requirements are completed in Question 2.
- Variables and all required data types are demonstrated in Question 3.
- All files are pushed to GitHub.
- GitHub repository link is ready to submit.

Happy Coding! ■